The Role of Artificial Intelligence With Deep Convolutional Neural Network in Screening Melanoma: A Systematic Review and Meta-Analyses of Quasi-Experimental Diagnostic Studies

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Introduction: Detecting melanoma as one of the most common skin cancer with using artificial intelligence (AI), such as deep convolutional neural network (DCNN) have the potency to increase the accuracy of the diagnosis. The aim of this study is to analyze the sensitivity, specificity, precision, and F1-score of DCNN in screening melanoma.

Methodology: The authors followed the PRISMA 2020 guidelines to retrieve literature in the following databases: PubMed, EBSCOhost, Emerald, Wiley, and ScienceDirect. The study's inclusion criteria were human quasi-experimental investigated DCNN in screening melanoma. The analysis was conducted using RevMan 5.4 and Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) to ensure the quality of the studies.

Results: Fifty-six of 2386 articles published in 2003 to 2023 were included and 24 studies were statistically analyzed. Various type of DCNN was used [artificial neural network (n = 4); pigment network (n = 4); atypical pigment network (n = 1); ResNet (= 8); AlexNet (n = 3); visual geometry group (n = 7); inception (n = 4); custom DCNN (n = 4)]. The mean and median of total sample size in meta-analysis with melanoma subjects were (18,791; 2,157) with (573; 261), respectively. Overall, QUADAS-2 showed low risk of bias. Diagnostic performance was observed with pooled sensitivity (0.881), pooled specificity (0.897), and pooled AUC (0.894). The precision and F1-score were ranging from 58% to 98.83% and 0.45 to 0.98. The forest plot and summary receiver operating characteristics curve (SROC) of each multiple in multiple analysis showed satisfactory results.

Conclusions: DCNN showed significant result to screen melanoma in patients. It has the potential to help clinician in giving early screening.

Key Words: Artificial intelligence, melanoma, screening, skin cancer

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This study was conducted in compliance with the principles of the Declaration of Helsinki. Consent was received from the patients.

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Melanoma is a cancer that originates from melanocytes. In the USA, melanoma affects 22.1 of 100,000 people annually (Cancer statistics from the Centers for Disease Control and Prevention). Melanoma has a high mortality rate; it causes 75% of skin cancer deaths despite representing only 4% of all skin cancer cases. In 2019, 96,480 new cases of melanoma were reported in the USA, among those, 7230 resulted in death.[1–3] Although histopathologic examination remains the gold standard, it is invasive. Noninvasive methods, such as the use of dermoscope, aids in the early diagnosis of melanoma and other variants of melanocytic intraepidermal lesions.[3,4] The use of image analysis technology using computer programs, such as deep convolutional networks (DCNN) and other artificial intelligence (AI), has the potential to increase the survival rate of patients with melanoma.[5–7]

Artificial Intelligence (AI), particularly DCNN, has emerged as a groundbreaking technology in medical diagnostics.

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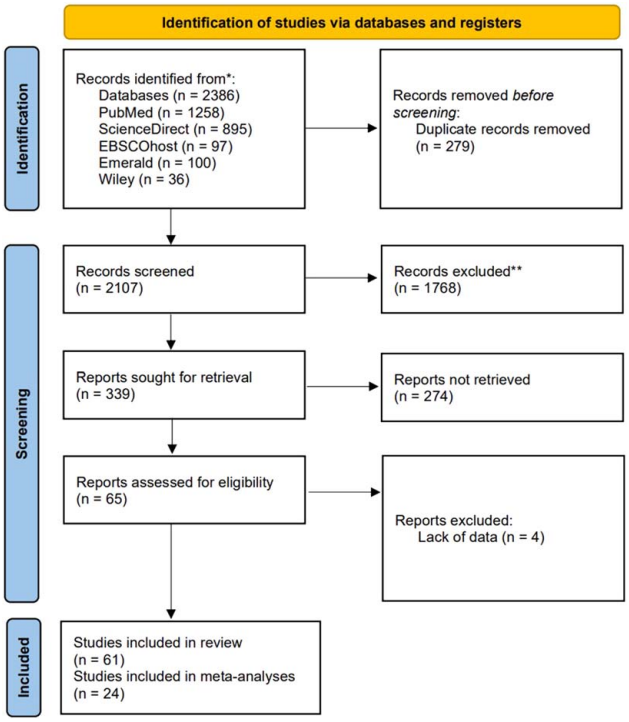


FIGURE 1. PRISMA flowchart 2020.

DCNNs are neural networks that are specially designed to analyze and process visual data. Unlike traditional image analysis techniques, DCNN utilizes multiple computational layers to extract complex features, making them highly effective for pattern recognition in medical imaging.[6] The unique architecture of DCNN enables the detection of intricate and subtle abnormalities, such as those present in dermoscopic images of melanoma.[7]

The combination of AI and DCNN technology has demonstrated remarkable potential in melanoma screening and diagnosis. Recent studies have demonstrated that DCNN-based models are comparable or even exceed the diagnostic accuracy of experienced dermatologists.[8] These models exhibit high sensitivity and specificity in identifying melanoma, effectively reducing false-negative rates and improving early detection.[9] Using automated image analysis, DCNN enhance clinical decision-making, allowing for faster and more objective assessments.[10]

Until now, no previously published meta-analysis on assessing the role of DCNN in diagnosing melanoma is available. The purpose of this paper is to determine the significance of DCNN in diagnosing melanoma. With faster and earlier diagnosis, it is hoped that therapy can be initiated quickly to reduce morbidity and mortality rates in melanoma patients.

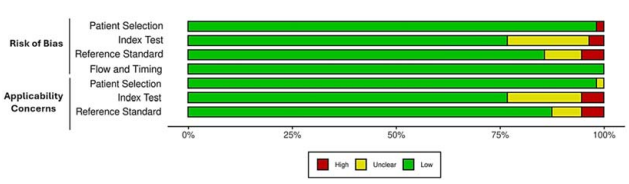


FIGURE 2. Traffic plot of QUADAS-2 risk of bias.

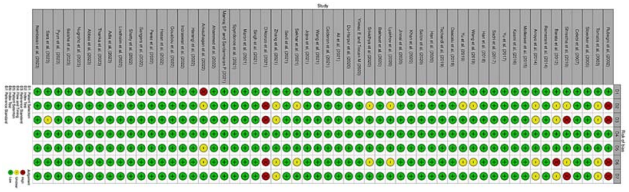


FIGURE 3. Summary plot of QUADAS-2 risk of bias.

METHODS

This study adhered to the Preferred Reporting Items guidelines for Systematic Reviews and Meta-analysis. Figure 1 shows the PRISMA 2020 checklist.

Literature Search

We conducted a comprehensive literature search to assess the potential of DCNN in screening melanoma from several databases: ProQuest, PubMed, EBSCOhost, SAGE, JSTOR, Emerald, Wiley, and ScienceDirect. The articles were retrieved with no time inception (until May 1, 2024). The search terms applied in PubMed are the following: (((((“skin”) AND ((“imaging”)) OR (“images”))) AND (“melanoma”)) OR ((“skin cancer”)) AND (“pigment*“)). A literature search was limited to those published in the English language and full-text. In addition, we manually searched additional possible studies for references.

Study Selection

Literature search was limited to articles published in English and full-text was available. Studies were included if they met the inclusion criteria: (1) written in English, (2) full-text is available, (3) investigated the potency of DCNN in diagnosing melanoma. Studies were excluded if they were: (1) Reviews, letter to editor, case report; (2) repeat publications; (3) lack of data availability.

Data Extraction

The authors extracted further information independently about the research that has been included, using Rayyan. The extracted data were as follows: (1) the name of the first author and publication year; (2) country; (3) study design; (4) subjects (number of melanoma, number of nonmelanoma); (5) methods (sources, devices); (6) neural network family; (7) outcomes (sensitivity, specificity, accuracy, area under the curve, precision, and F1-scores).

Quality Assessment

Seven independent reviewers assessed the bias risk by applying the Quality Assessment of Diagnostic Accuracy Studies-2

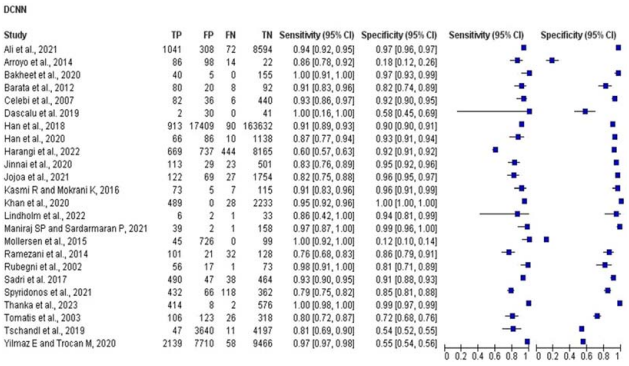


FIGURE 4. Forest plot of DCNN diagnostic performance.

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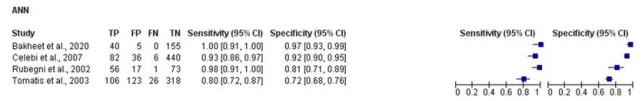


FIGURE 5. Forest plot of ANN diagnostic performance.

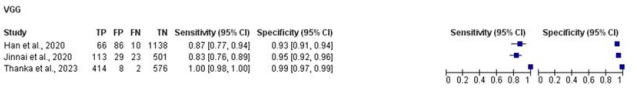


FIGURE 7. Forest plot of VGG diagnostic performance.

(QUADAS-2) risk of bias (RoB) tool. Any disagreements will be resolved through discussion by the corresponding author.

Statistical Analysis Outcome Measurement

The outcome of this review was to analyze and evaluate the accuracy, sensitivity, specificity, area under curve, precision, and F1-score. The included studies did not directly employ the use of human or animal subjects. Each study was conducted in compliance with the ethical standards specified therein. Statistical analysis was conducted using RevMan 5.4. Multiple analysis was conducted for Artificial Neural Network (ANN), ResNet, Visual Geometry Group Network (VGG), Pigment Network, GoogleNet, and custom.

RESULTS

A total of 2386 articles were obtained from literature search of databases. After the removal of duplicates and screening, 61 articles were included in the systematic review and 24 articles were included in the meta-analysis. The PRISMA 2020 flowchart can be seen in Figure 1. The risk of bias of included studies was then assessed using the QUADAS-2 for diagnostic studies. It was found that the majority of studies had a low risk of bias (Figs. 2 and 3).

From the systematic review of 61 studies (Supplement Table 1, Supplemental Digital Content 1, http://links.lww.com/SCS/H876), it was found that the sensitivity ranged from 38% to 100% with a mean of 84.95% and a pooled (0.881). Specificity ranged from 53.5% to 100% with a mean of 86.89% and pooled (0.897). Accuracy ranged from 74% to 99.1% with a mean of 89.93%, and pooled AUC (0.894). Meanwhile, precision and F1-score ranged from 58% to 98.83% and 45% to 98%, respectively, with a mean precision of 80.49% and a mean F1-score of 78.18%.

The lowest sensitivity (38%) was obtained in the study by Sevil et al,[11] whereas the highest (100%) was in research observed in the study by Bakheet et al[12] Meanwhile, the lowest specificity (53.5%) was obtained in the research by Tschandl et al[13] and the highest (100%) in the research conducted by Khan et al.[10] In terms of accuracy, the lowest value (74%) was obtained in the research by Tomatis et al[14] and the highest value (99.1%) was obtained in the study by Roshni Thanka et al[15] The highest precision value and F1-score (each 98.83%) were obtained in the research of Balaha et al.[8] Through statistical analysis of the 24 studies selected for meta-analysis, sensitivity was shown to range from 0.76 to 1.00, with the majority >0.85. Specificity varied more widely, from 0.12 to 1.00. The DCNN model showed high sensitivity, but some studies have noted difficulty in correctly identifying nonmelanoma (true negative) cases as seen in Figure 4. In the multiple analysis of artificial neural network (ANN) (n = 4), sensitivity ranged from 0.72 to 1.00. Specificity ranged from 0.72 to 0.97. ANN showed high sensitivity and relatively balanced ability to correctly classify nonmelanoma cases, compared with DCNN, as seen in Figure 5. In the ResNet (n = 4) multiple analysis, sensitivity ranged from 0.79 to 0.91. Specificity ranged from 0.85 to 0.90. The ResNet model performs well in terms of sensitivity, with high specificity as seen in Figure 6. In the visual geometry group network (VGG; n = 3) multiple, sensitivity ranged from 0.87 to 1.00. Specificity ranged from 0.93 to 0.99. VGG showed high sensitivity, with consistent specificity as seen in Figure 7. In the multiple analysis of the pigment network (n = 4), sensitivity ranged from 0.86 to 1.00. Specificity ranged from 0.12 to 0.96. Pigment Network showed high sensitivity, but lower specificity compared with other models as seen in Figure 8. In the GoogleNet multiple, sensitivity ranged from 0.93 to 1.00. Specificity ranged from 0.55 to 1.00. GoogleNet showed high sensitivity, with lower specificity compared with other models as seen in Figure 9. In the custom multiple, sensitivity showed a high range, from 0.86 to 0.94. Specificity ranged between 0.91 and 0.94. Custom networks showed high sensitivity and specificity as seen in Figure 10.

The majority of models (especially DCNN, ANN, and Custom) were clustered in the upper left portion of the plot, indicating high sensitivity (>0.80) and high specificity (around 0.80–1.00). These models were significantly effective in screening melanoma while maintaining accuracy in identifying nonmelanoma cases. By plotting the data into the summary receiver operating characteristics (SROC) curve, it was found that all 24 studies were located above the curve. Models using DCNN, ANN, and custom were found to be located in the upper left portion of the plot (Fig. 11).

Sensitivity ranging from 0.71 to 1.00. Specificity Ranging from 0.12 to 1.00. DCNN with dermoscopy showed high sensitivity, with various specificity. Another multiple analysis was carried out on the sensitivity and specificity of combining DCNN with dermoscopy (n = 22). Sensitivity was found to range from 0.71 to 1.00, specificity ranged from 0.12 to 1.00. Combination of DCNN with dermoscopy shows a high sensitivity, but heterogenous results of specificity (Fig. 12).

The majority of data points were clustered in the upper left quadrant of the plot, with high sensitivity (0.80–1.00) and high specificity (0.70–1.00). DCNN with dermoscopy were highly sensitive in screening true melanoma cases (low false negatives). Specificity was also generally high, but with some variability across the data points. By plotting the data from the combination of DCNN and dermoscopy into the summary receiver operating characteristics (SROC) curve, it was found that all 22 studies were located above the curve (Fig. 13).

As indicated in Supplement Tables 2 and 3, Supplemental Digital Content 1, http://links.lww.com/SCS/H876, the majority of the included studies' GRADE Quality of Evidence Assessments indicated moderate to high evidence quality. Small sample sizes and variations in study designs are the main

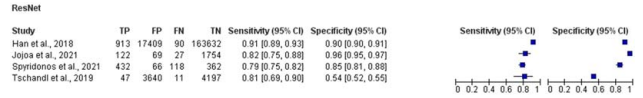


FIGURE 6. Forest plot of ResNet diagnostic performance.

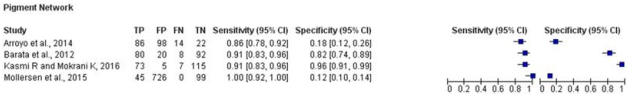


FIGURE 8. Forest plot of PN diagnostic performance.

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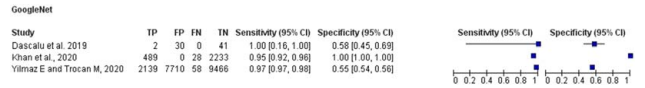


FIGURE 9. Forest plot of GoogleNet diagnostic performance.

drawbacks. However, despite minor imprecision caused by wide CI, our study provides valuable insight into melanoma screening in the clinical setting.

DISCUSSION

This systematic review and meta-analysis highlighted the direct comparison of DCNN in diagnosing and screening melanoma between available studies, from methods to efficacy. To enhance the quality of the systematic review, we conducted a statistical analysis to draw a conclusion.

The overall outcome of using DCNN in machine learning was satisfactory in distinguishing melanoma in patients. Our results found pooled sensitivity (0.881), pooled specificity (0.897), and pooled AUC (0.894). The precision and F1-score ranged from 58% to 98.83% and 0.45 to 0.98, respectively. The findings were in line with the research conducted by Patel and colleagues, which aimed to assess the current usage of AI-based methods alongside noninvasive diagnostic imaging techniques for melanoma detection. AI algorithms consistently demonstrated robust performance, with an average ROC (receiver operating characteristic) exceeding 80% in identifying melanoma. In analyses using dermoscopic images, the average sensitivity and specificity of the algorithms were ~83.01% and 85.58%, respectively.[9]

The DCNN represents a specialized category of artificial neural networks (ANN) that utilize convolution instead of general matrix multiplication in at least one layer. Unlike simple neural networks with limited hidden layers, CNNs are composed of numerous layers, enabling them to effectively capture complex and nonlinear functions. CNN architecture typically features various types of layers, such as convolutional, pooling, and fully connected layers, incorporating regularization techniques. To comprehend intricate features and functions necessary for representing high-level abstractions in tasks like vision and language processing, CNNs require deep architectures. These deep architectures, including CNNs, are characterized by a large number of neurons and multiple layers facilitating the computation of nonlinear transformations.[6]

Despite the promising results of DCNN, particularly when combined with several devices, our study has a number of drawbacks. The absence of a standardization procedure among protocols could result in false positives and false negatives. Varied devices between studies and small sample sizes may possibly have an impact on the study's findings. Randomized controlled trials are needed to create a standardized methodology for the diagnostic protocols for DCNN in screening melanoma.

Although DCNN models have demonstrated diagnostic performance comparable to, or in some cases surpassing, that of experienced dermatologists, we believe these systems are best positioned as augmented intelligence tools, rather than replacements for clinicians. Their role is to support and enhance

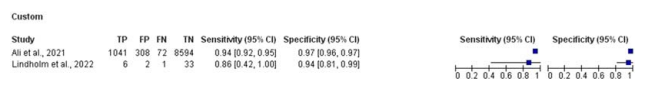


FIGURE 10. Forest plot of custom DCNN diagnostic performance.

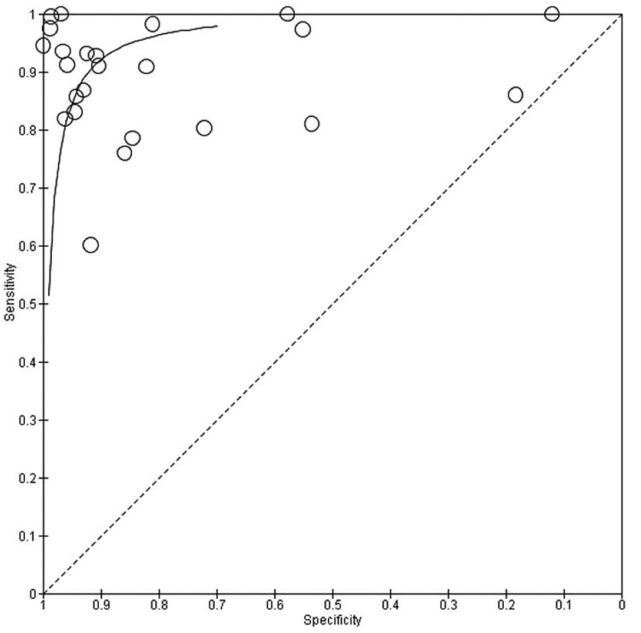


FIGURE 11. Summary receiver operating characteristic (SROC) curve of the included studies.

decision-making by providing rapid, objective assessments of dermoscopic images, especially in settings with limited access to dermatology specialists. Ultimately, clinical judgment remains irreplaceable, as patient history, physical examination, and holistic evaluation are essential components of diagnosis and management. Therefore, DCNN should be viewed as assistive technology that complements—not supplants—human expertise.[15–32]

The medico-legal implications of AI-assisted diagnosis remain an evolving area. Current guidelines suggest that clinicians remain responsible for final diagnostic decisions, even when using AI tools. Regulatory bodies, such as the FDA and EMA, have begun issuing frameworks for Software as a Medical Device (SaMD), emphasizing transparency, validation, and traceability. In the event of a misdiagnosis, liability typically depends on how the AI system was used (eg, as a decision aid versus autonomous decision-maker), whether it met regulatory standards, and whether the clinician adhered to best practices. Further legal clarity is needed, but ethical deployment requires proper validation, informed consent, and clinician oversight.[32–48]

The implementation costs of DCNN technology vary depending on the system's complexity, integration with existing electronic health records, and whether it is cloud-based or locally hosted. Open-source models (eg, those trained using public

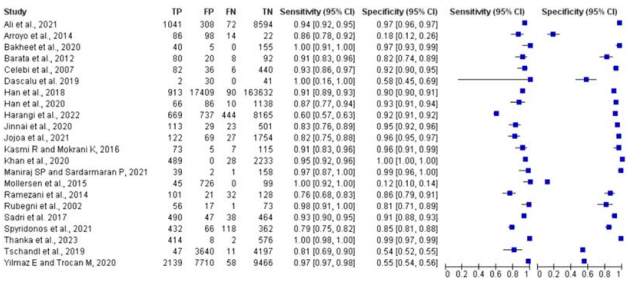


FIGURE 12. Forest plot of DCNN with dermoscopy.

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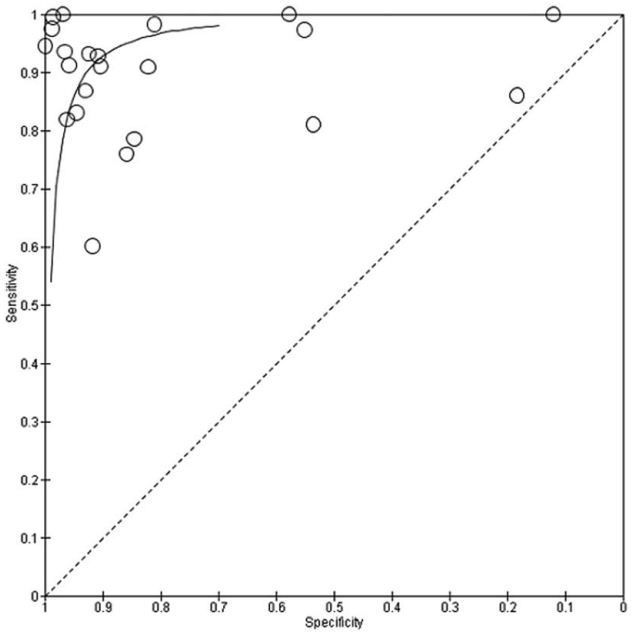


FIGURE 13. Summary receiver operating characteristic (SROC) curve of DCNN with dermoscopy.

data sets like ISIC) offer a low-cost entry point but still require infrastructure and expertise for deployment. Commercial solutions can involve licensing fees, typically ranging from a few thousand to tens of thousands of dollars annually, depending on scale. In terms of time, once deployed, DCNN systems provide near-instantaneous image analysis, significantly reducing the time burden on clinicians. Cost-benefit analyses suggest that, when applied appropriately, AI systems may reduce unnecessary biopsies, improve triage, and optimize dermatology referrals, thereby offering long-term savings.[48–62]

CONCLUSION

From our study, we can conclude that applying DCNN in machine learning was satisfactory in screening melanoma in patients. DCNN, with and without a dermoscope, has the potential to diagnose skin cancer. Thus, DCCN has the potential to aid clinicians in early screening and diagnosing, allowing patients to quickly initiate treatment. These studies have become a big stone in the development of diagnosis techniques. As for now, the usage of AI is limited to focusing on the appearance of skin lesions. Therefore, clinicians still need to approach patients holistically to obtain the full diagnosis.

REFERENCES

1. Davis LE, Shalin SC, Tackett AJ. Current state of melanoma diagnosis and treatment. *Cancer Biol Ther* 2019;20:1366–1379
2. Centers for Disease Control and Prevention (CDC). Cancer Data and Statistics. 2024. http://www.cdc.gov/cancer/data/index.html
3. Piłat P, Borzęcki A, Jazienicki M, et al. Skin melanoma imaging using ultrasonography: a literature review. *Adv Dermatol Allergol* 2018;35:238–242
4. Dinnes J, Deeks JJ, Chuchu N, et al. Dermoscopy, with and without visual inspection, for diagnosing melanoma in adults. *Cochrane Database Syst Rev* 2018;12:CD011902
5. Tajerian A, Kazemian M, Tajerian M, et al. Design and validation of a new machine-learning-based diagnostic tool for the differentiation of dermatoscopic skin cancer images. *PLoS One* 2023;18:e0284437
6. Namatēvs I. Deep convolutional neural networks: structure, feature extraction and training. *Inf Technol Manage Sci* 2017;20:23–27
7. Arivazhagan N, Mukunthan MA, Sundaranarayana D, et al. Analysis of skin cancer and patient healthcare using data mining techniques. *Comput Intell Neurosci* 2022;2022:2250275
8. Balaha HM, Hassan AES. Skin cancer diagnosis based on deep transfer learning and sparrow search algorithm. *Neural Comput Appl* 2023;35:815–853
9. Patel RH, Foltz EA, Witkowski A, et al. Analysis of artificial intelligence-based approaches applied to non-invasive imaging for early detection of melanoma: a systematic review. *Cancers (Basel)* 2023;15:4694
10. Khan MA, Akram T, Sharif M, et al. An integrated framework of skin lesion detection and recognition through saliency method and optimal deep neural network features selection. *Neural Comput Appl* 2020;32:15929–15948
11. Sevli O. A deep convolutional neural network-based pigmented skin lesion classification application and experts evaluation. *Neural Comput Appl* 2021;33:12039–12050
12. Bakheet S, Al-Hamadi A. Computer-aided diagnosis of malignant melanoma using Gabor-based entropic features and multilevel neural networks. *Diagnostics* 2020;10:822
13. Tschandl P, Rosendahl C, Akay BN, et al. Expert-level diagnosis of nonpigmented skin cancer by combined convolutional neural networks. *JAMA Dermatol* 2019;155:58
14. Tomatis S, Bono A, Bartoli C, et al. Automated melanoma detection: multispectral imaging and neural network approach for classification. *Med Phys* 2003;30:212–221
15. Roshni Thanka M, Bijolin Edwin E, Ebenezer V, et al. A hybrid approach for melanoma classification using ensemble machine learning techniques with deep transfer learning. *Comput Methods Prog Biomed Update* 2023;3:100103
16. Celebi ME, Kingravi HA, Uddin B, et al. A methodological approach to the classification of dermoscopy images. *Comput Med Imaging Graph* 2007;31:362–373
17. Rubegni P, Burroni M, Perotti R, et al. Digital dermoscopy analysis and artificial neural network for the differentiation of clinically atypical pigmented skin lesions: a retrospective study. *J Investig Dermatol* 2002;119:471–474
18. Han SS, Kim MS, Lim W, et al. Classification of the clinical images for benign and malignant cutaneous tumors using a deep learning algorithm. *J Investig Dermatol* 2018;138:1529–1538
19. Yu Z, Jiang X, Zhou F, et al. Melanoma recognition in dermoscopy images via aggregated deep convolutional features. *IEEE Trans Biomed Eng* 2019;66:1006–1016
20. Han SS, Park I, Eun Chang S, et al. Augmented intelligence dermatology: deep neural networks empower medical professionals in diagnosing skin cancer and predicting treatment options for 134 skin disorders. *J Investig Dermatol* 2020;140:1753–1761
21. Saitov I, Polevaya T, Filchenkov A. Dermoscopic attributes classification using deep learning and multi-task learning. *Procedia Comput Sci* 2020;178:328–336
22. Lyakhov PA, Lyakhova UA, Nagornov NN. System for the recognizing of pigmented skin lesions with fusion and analysis of heterogeneous data based on a multimodal neural network. *Cancers (Basel)* 2022;14:1819
23. Jojoa Acosta MF, Caballero Tovar LY, Garcia-Zapirain MB, et al. Melanoma diagnosis using deep learning techniques on dermatoscopic images. *BMC Med Imaging* 2021;21:6
24. Spyridonos P, Gaitanis G, Likas A, et al. Characterizing malignant melanoma clinically resembling seborrheic keratosis using deep knowledge transfer. *Cancers (Basel)* 2021;13:6300
25. Zhang B, Wang Z, Gao J, et al. Short-term lesion change detection for melanoma screening with novel siamese neural network. *IEEE Trans Med Imaging* 2021;40:840–851
26. Maron RC, Haggenmüller S, von Kalle C, et al. Robustness of convolutional neural networks in recognition of pigmented skin lesions. *Eur J Cancer* 2021;145:81–91
27. Jinnai S, Yamazaki N, Hirano Y, et al. The development of a skin cancer classification system for pigmented skin lesions using deep learning. *Biomolecules* 2020;10:1123

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28. Shrestha B, Bishop J, Kam K, et al. Detection of atypical texture features in early malignant melanoma. *Skin Res Technol* 2010;16: 60–65
29. Barata C, Marques JS, Rozeira J. A system for the detection of pigment network in dermoscopy images using directional filters. *IEEE Trans Biomed Eng* 2012;59:2744–2754
30. García Arroyo JL, García Zapirain B. Detection of pigment network in dermoscopy images using supervised machine learning and structural analysis. *Comput Biol Med* 2014;44:144–157
31. Kasmi R, Mokrani K. Classification of malignant melanoma and benign skin lesions: implementation of automatic ABCD rule. *IET Image Process* 2016;10:448–455
32. Møllersen K, Kirchesch H, Zortea M, et al. Computer-aided decision support for melanoma detection applied on melanocytic and nonmelanocytic skin lesions: a comparison of two systems based on automatic analysis of dermoscopic images. *Biomed Res Int* 2015;2015:579282
33. Dascalu A, David EO. Skin cancer detection by deep learning and sound analysis algorithms: a prospective clinical study of an elementary dermoscope. *EBioMedicine* 2019;43:107–113
34. Yilmaz E, Trocan M. A modified version of GoogLeNet for melanoma diagnosis. *J Inf Telecommun* 2021;5:395–405
35. Nugroho ES, Ardiyanto I, Nugroho HA. Boosting the performance of pretrained CNN architecture on dermoscopic pigmented skin lesion classification. *Skin Res Technol* 2023;29:22–48
36. Du-Harpur X, Arthurs C, Ganier C, et al. Clinically relevant vulnerabilities of deep machine learning systems for skin cancer diagnosis. *J Investig Dermatol* 2021;141:916–920
37. Ali MS, Miah MS, Haque J, et al. An enhanced technique of skin cancer classification using deep convolutional neural network with transfer learning models. *Machine Learn Appl* 2021;5:100036
38. Lindholm V, Raita-Hakola AM, Annala L, et al. Differentiating malignant from benign pigmented or non-pigmented skin tumours —a pilot study on 3D hyperspectral imaging of complex skin surfaces and convolutional neural networks. *J Clin Med* 2022;11: 1914
39. Stoecker WV, Gupta K, Stanley RJ, et al. Detection of asymmetric blotches (asymmetric structureless areas) in dermoscopy images of malignant melanoma using relative color. *Skin Res Technol* 2005; 11:179–184
40. Ramezani M, Karimian A, Moallem P. Automatic detection of malignant melanoma using macroscopic images. *J Med Signals Sens* 2014;4:281
41. Sadri AR, Azarianpour S, Zekri M, et al. WN-based approach to melanoma diagnosis from dermoscopy images. *IET Image Process* 2017;11:475–482
42. Srividhya V, Sujatha K, Ponmagal RS, et al. Vision based detection and categorization of skin lesions using deep learning neural networks. *Procedia Comput Sci* 2020;171:1726–1735
43. Calderón C, Sanchez K, Castillo S, et al. BILSK: a bilinear convolutional neural network approach for skin lesion classification. *Comput Methods Prog Biomed Update* 2021;1:100036
44. Wang S, Hamian M. Skin cancer detection based on extreme learning machine and a developed version of thermal exchange optimization. *Comput Intell Neurosci* 2021;2021:64–73
45. Sankar Raja Sekhar K, Ranga Babu T, Prathibha G, et al. Dermoscopic image classification using CNN with Handcrafted features. *J King Saud Univ Sci* 2021;33:101550
46. D'Alonzo M, Bozkurt A, Alessi-Fox C, et al. Semantic segmentation of reflectance confocal microscopy mosaics of pigmented lesions using weak labels. *Sci Rep* 2021;11:3679
47. Singh L, Janghel RR, Sahu SP. TrCSVM: a novel approach for the classification of melanoma skin cancer using transfer learning. *Data Technol Appl* 2020;55:64–81
48. Maniraj SP, Sardarmaran P. Classification of dermoscopic images using soft computing techniques. *Neural Comput Appl* 2021;33: 13015–13026
49. Ahammed M, Mamun Md Al, Uddin MS. A machine learning approach for skin disease detection and classification using image segmentation. *Healthcare Analytics* 2022;2:100122
50. Indraswari R, Rokhana R, Herulambang W. Melanoma image classification based on MobileNetV2 network. *Procedia Comput Sci* 2022;197:198–207
51. Foahom Gouabou AC, Collenne J, Monnier J, et al. Computer aided diagnosis of melanoma using deep neural networks and game theory: application on dermoscopic images of skin lesions. *Int J Mol Sci* 2022;23:13838
52. Yu L, Chen H, Dou Q, et al. Automated melanoma recognition in dermoscopy images via very deep residual networks. *IEEE Trans Med Imaging* 2017;36:994–1004
53. Wang X, Jiang X, Ding H, et al. Bi-directional dermoscopic feature learning and multi-scale consistent decision fusion for skin lesion segmentation. *IEEE Trans Image Process* 2020;29:3039–3051
54. Hasan MdK, Elahi MdTE, Alam MdA, et al. DermoExpert: skin lesion classification using a hybrid convolutional neural network through segmentation, transfer learning, and augmentation. *Inform Med Unlocked* 2022;28:100819
55. Pérez E, Ventura S. An ensemble-based convolutional neural network model powered by a genetic algorithm for melanoma diagnosis. *Neural Comput Appl* 2022;34:10429–10448
56. Sangers T, Reeder S, van der Vet S, et al. Validation of a market-approved artificial intelligence mobile health app for skin cancer screening: a prospective multicenter diagnostic accuracy study. *Dermatology* 2022;238:649–656
57. Shetty B, Fernandes R, Rodrigues AP, et al. Skin lesion classification of dermoscopic images using machine learning and convolutional neural network. *Sci Rep* 2022;12:18134
58. Adla D, Reddy GVR, Nayak P, et al. A full-resolution convolutional network with a dynamic graph cut algorithm for skin cancer classification and detection. *Healthcare Analytics* 2023;3: 100154
59. Abbas Q, Daadaa Y, Rashid U, et al. Assist-Dermo: a lightweight separable vision transformer model for multiclass skin lesion classification. *Diagnostics* 2023;13:2531
60. Pyun SH, Min W, Goo B, et al. Real-time, in vivo skin cancer triage by laser-induced plasma spectroscopy combined with a deep learning–based diagnostic algorithm. *J Am Acad Dermatol* 2023;89: 99–105
61. Cerminara SE, Cheng P, Kostner L, et al. Diagnostic performance of augmented intelligence with 2D and 3D total body photography and convolutional neural networks in a high-risk population for melanoma under real-world conditions: a new era of skin cancer screening? *Eur J Cancer* 2023;190:112954
62. Nambisan AK, Maurya A, Lama N, et al. Improving automatic melanoma diagnosis using deep learning-based segmentation of irregular networks. *Cancers (Basel)* 2023;15:1259

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